Experiment 4
LAB REPORT

Data Modelling and Form Design for a Business Application in Zoho Creator

Project Name: Library Book Issue

Author: Pavithran M
Roll No: 241401065

241401065

Data Modelling and Form Design for a Business Application in Zoho Creator

Date : 17th August 2026

AIM

To design and build a “Library Book Issue” application in Zoho Creator that demonstrates correct usage of all thirteen Basic Fields available in the Form Builder — Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Multi Select, and Checkbox — and to model the underlying data structure of a real fitness-center enrollment workflow through a properly sectioned form and an auto-generated report view.

PROCEDURE

1. Data Model Design

Before building the form, the data requirement was analysed and organised into a simple data model, grouping the thirteen Basic Fields under four logical sections that map to the stages of a member’s enrollment journey: Member Details, Contact Details, Membership & Fitness Preferences, and Scheduling & Health Declaration. Every field submitted through the form flows into a single auto-generated report, Membership_Enrollment_Report, which the front desk and trainers use to track and manage enrollments. The resulting data model is shown below.

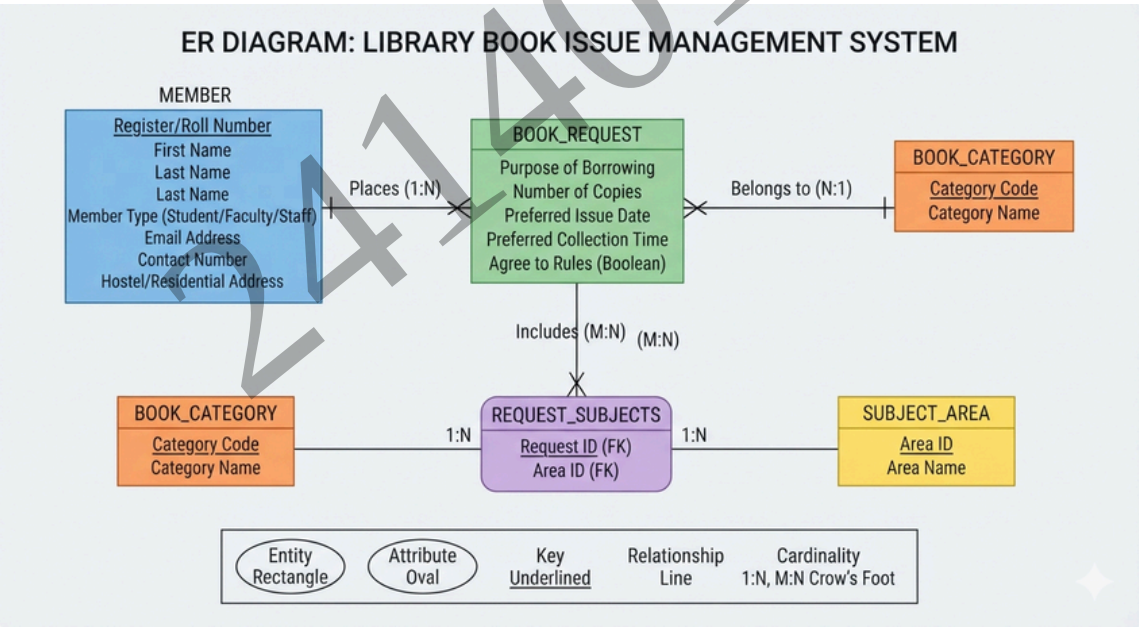


Fig 1. Data Model — Membership_Enrollment_Form and its sectioned fields feeding Membership_Enrollment_Report

2. Application and Form Creation

1. Logged into Zoho Creator and clicked Create New Application → Create from Scratch.
2. Named the application “Gym Membership Enrollment & Front Desk Management” and confirmed creation, which opened the default blank workspace.
3. Inside the application, opened the Forms panel and clicked the + icon to add a new form, naming it Membership_Enrollment_Form.
4. This automatically created a linked report, Membership_Enrollment_Report, which will list every submitted enrollment request.

3. Drag-and-Drop of Basic Fields

In the Form Builder, the Basic Fields palette was expanded on the left panel. Each required field was dragged from the palette and dropped onto the canvas in the order needed, then double-clicked to open its Field Properties panel for renaming and configuration, as listed below.

Basic Field	Form Field Name	Purpose	Sample Value
Phone	Contact_Number	Captures a validated mobile number for the library to notify the member when the book is ready	+91 97890 55667
Single Line	Register_Roll_Number	Captures a short free-text entry for the member's register/roll/employee number	22ADR045
Multi Line	Purpose_Of_Borrowing	Captures free-text description of why the book is needed, e.g. project or reference use	Required as reference material for the final-year capstone project on AI-based automation.
Number	Number_Of_Copies	Captures a numeric value with validation for the number of books being requested	2
Date	Preferred_Issue_Date	Captures the member's preferred date to collect the book, using a date picker	28-Jul-2026
Time	Preferred_Collection_Time	Captures the member's preferred time slot to visit the counter and collect the book	2:00 PM
Drop Down	Book_Category	Presents a single-select list of library sections/categories the book belongs to	Computer Science & Engineering
Radio	Member_Type	Captures a single mandatory selection among mutually exclusive membership types	Student
Multi Select	Subject_Areas_Of_Interest	Allows selection of one or more subject areas for which similar books may be recommended	Artificial Intelligence, Data Science, Machine Learning
Checkbox	Agree_To_Library_Rules	Captures the member's acknowledgement of library rules and the commitment to return the book by the due date	Checked (Agreed)

2415065

4. Organising Fields into Sections

To keep the form member-friendly and logically grouped, a Section field was dragged from the Layout palette above each group of related fields, and the dropped fields were dragged again to nest them under the correct section header. Four sections were created in this way:

- Section A — Member Details: Member Name, Age, Membership Type.
- Section B — Contact Details: Email Address, Contact Number, Residential Address, Emergency Contact Name.
- Section C — Membership & Fitness Preferences: Membership Plan, Preferred Activities, Fitness Goals / Medical Notes.
- Section D — Scheduling & Health Declaration: Preferred Start Date, Preferred Workout Slot, Health Declaration Waiver.

Each section header was renamed by double-clicking it and typing the section title, and fields within a section were reordered using drag handles until the layout matched the intended member journey from identity to health declaration.

5. Applying Field-Level Validations

1. Selected Member_Name, Contact_Number, Age, Membership_Type, Membership_Plan, and Health_Declaration_Waiver one by one and enabled the Mandatory toggle in Field Properties.
2. Opened Email_Address properties and confirmed the built-in email format validation was active.
3. Opened Contact_Number properties and set the phone format to the Indian country-code pattern.
4. Opened Age properties and set Minimum Value = 14 and Maximum Value = 75.
5. Opened Preferred_Start_Date properties and restricted selectable dates to today and future dates only, disallowing past dates.
6. Confirmed under Health_Declaration_Waiver that the Submit button remains disabled until the checkbox is checked.

6. Preview and Deployment

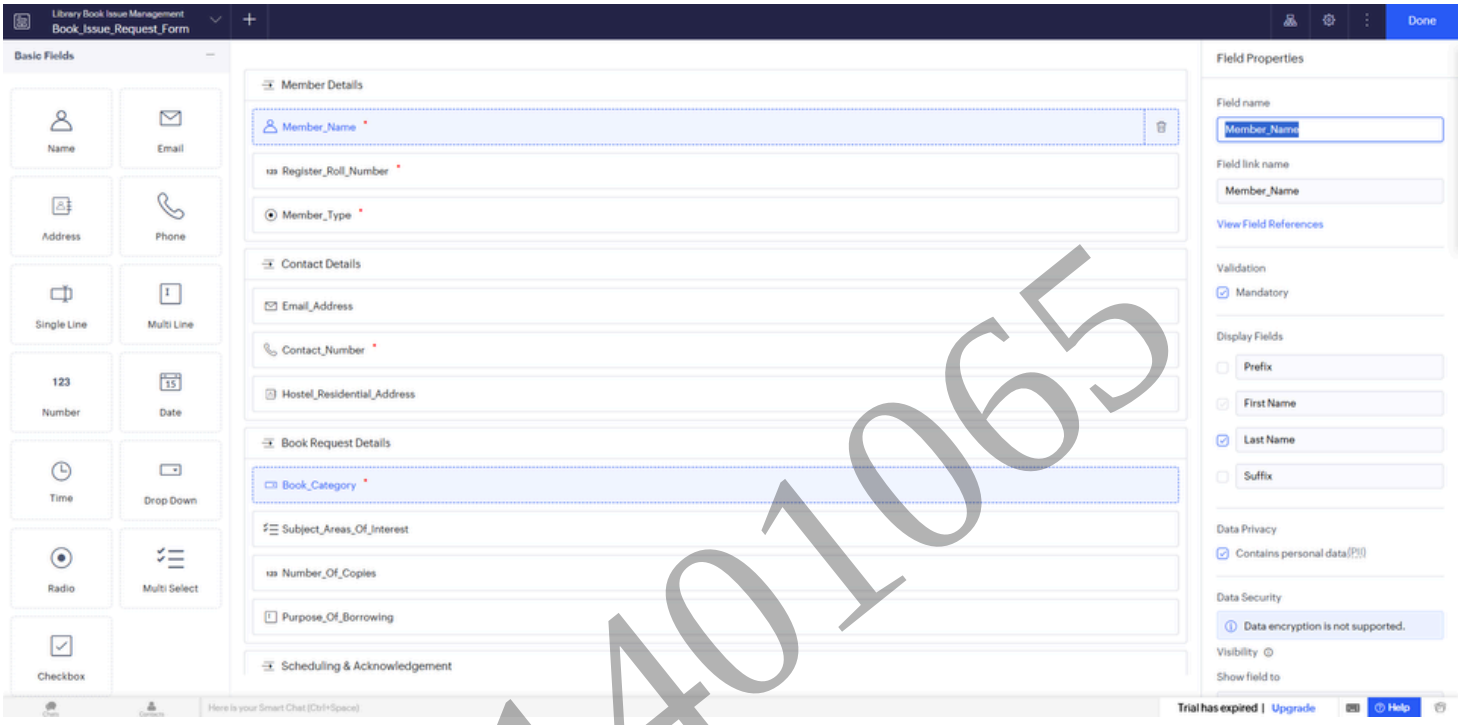
1. Clicked the Preview icon in the Form Builder toolbar to view the live member-facing form and verified that all four sections render in order with correct field types.
2. Filled and submitted a sample enrollment through the preview/live form to test validations and generate report data.
3. Opened Membership_Enrollment_Report from the Reports panel and verified that the submitted record appears with values mapped correctly to each field.
4. Reviewed filter options on Membership Plan and Preferred Start Date in the report view to confirm the front desk can sort and onboard new members.

24140105

OUTPUT

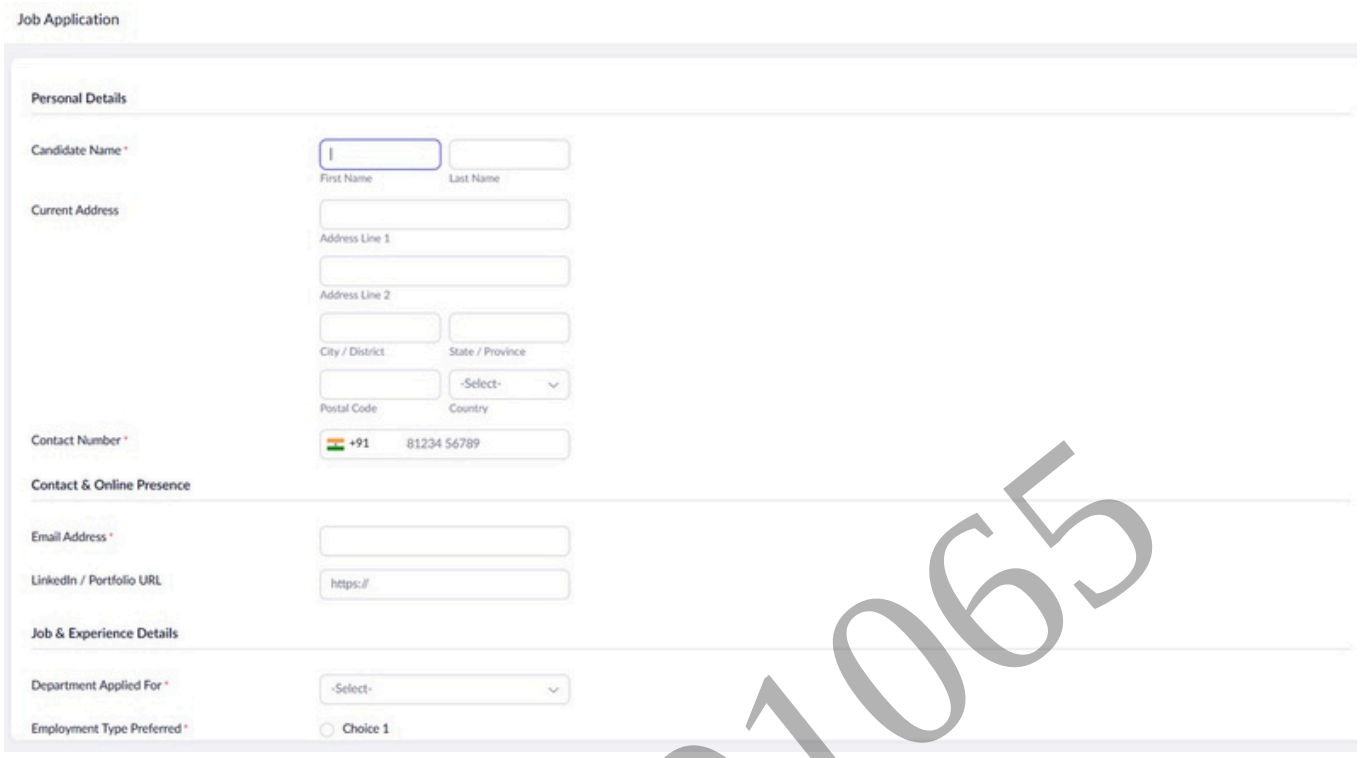
Thefollowing screenshots capture the completed application: the Form Builder showing all thirteen Basic Fields arranged into four sections, the live member-facing form preview, and the Membership_Enrollment_Report containing a sample submitted record.

1. FormBuilder View—All BasicFieldsPlaced and Sectioned



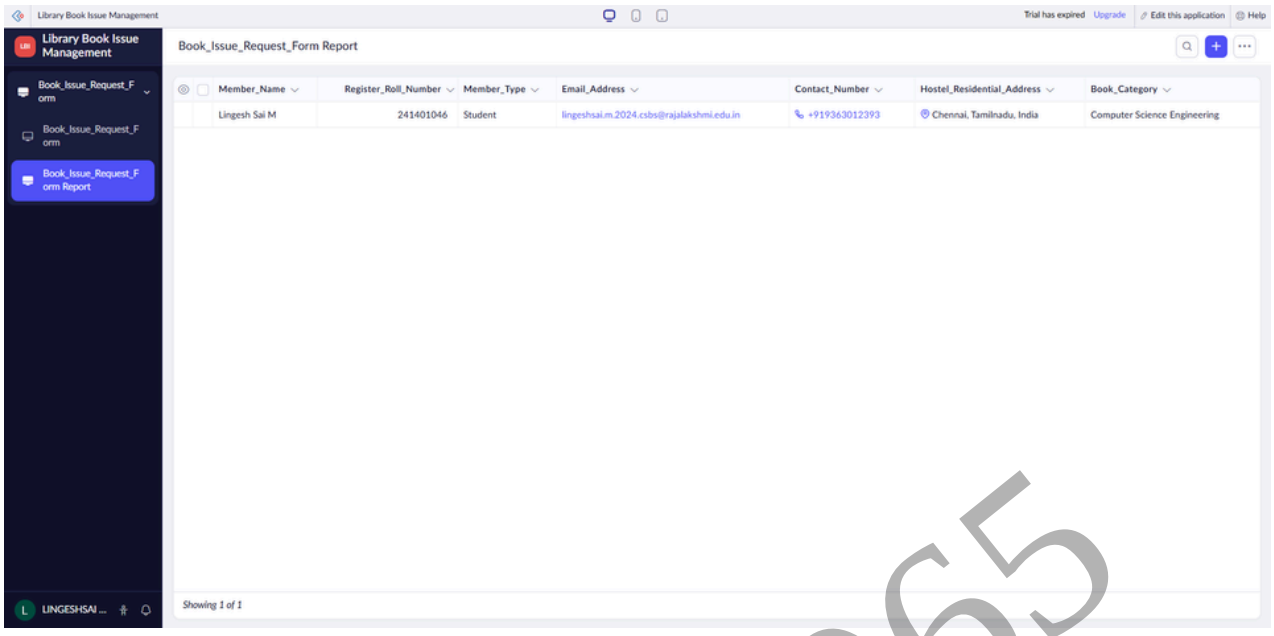
24140106 5

2. Live Form Preview



241401065

3. Membership_Enrollment_Report — Sample Submitted Record



RESULT

A Job Application Form application was successfully designed and built in Zoho Creator using all thirteen Basic Fields — Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, Drop Down, Radio, Multi Select, and Checkbox. The fields were logically grouped into four sections through drag-and-drop, mandatory and format-level validations were configured successfully, and the Membership_Enrollment_Report was verified to correctly capture and list member enrollment data submitted through the form. The exercise thus demonstrated an understanding of data modelling and low-code form design for a real business use case.

241401065